STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 9: Mechanical Properties of Fluids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define pressure. [1]
2. SI unit of dynamic viscosity? [1]
3. Continuity relation for incompressible flow? [1]
4. Bernoulli relation? [1]
5. Surface tension unit? [1]
6. Pressure rise at depth h? [1]
7. Stokes drag? [1]
8. Excess pressure in a liquid drop? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. Water at 3m depth, rho=1000,g=10: gauge pressure? [2]
10. If pipe area halves, incompressible speed? [2]
11. Under Bernoulli assumptions, faster horizontal flow means pressure? [2]
12. Force from p=5000 Pa over area .02 m^2? [2]
13. Drop S=.06 N/m,r=.003 m: excess pressure? [2]
14. Why is a soap bubble pressure difference twice a drop's? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. Pipe area 4 cm^2 at 2 m/s narrows to 1 cm^2. New speed? [3]
16. Horizontal water speeds 2 and 6 m/s, rho=1000. Find p_1-p_2. [3]
17. Stokes drag for r=.5 mm,=1 Pa s,v=.02 m/s? [3]
18. Explain why pressure rises with depth. [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Areas 6 cm^2 and 2 cm^2; inlet speed 1.5 m/s. Find outlet speed. [4]
20. Soap bubble r=2mm,S=.03N/m: excess pressure? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations and relations

- Pressure $p=F/A$; hydrostatic gauge pressure $p_g=\rho g h$; absolute pressure $p=p_0+\rho g h$.
- Continuity for incompressible flow: $A_1v_1=A_2v_2$.
- Bernoulli along a streamline: $p+(1/2)\rho v^2+\rho gh=\text{constant}$.
- Stokes' drag magnitude $F_d=6\pi \eta r v$ for a small sphere in laminar flow.
- Liquid drop excess pressure $\Delta p=2S/r$; soap bubble $\Delta p=4S/r$ (two surfaces).
- Flow speed ratio: $v_2/v_1=A_1/A_2$ for the same volume flow rate.
- Bernoulli at equal heights: $p_1-p_2=(1/2)\rho(v_2^2-v_1^2)$.
- Force from uniform pressure: $F=pA$; surface tension unit N/m.

Use equations only when their assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Concept questions are answered directly; numerical problems show the governing equation and substitution.

Question 1

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Normal force per unit area.
Final answer: Normal force per unit area.

Question 2

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Pa s.
Final answer: Pa s.

Question 3

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: A_1v_1=A_2v_2.
Final answer: A_1v_1=A_2v_2.

Question 4

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: p+rhov^2+rhogh=constant.
Final answer: p+rhov^2+rhogh=constant.

Question 5

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: N/m.
Final answer: N/m.

Question 6

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: rhogh.
Final answer: rhogh.

Question 7

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 6pirv.
Final answer: 6pirv.

Question 8

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 2S/r.
Final answer: 2S/r.

Question 9

Working: Gauge pressure p=rho gh=1000(10)(3)=30,000 Pa. This is pressure above atmospheric pressure.
Final answer: 30,000 Pa.

Question 10

Working: Continuity gives A_1v_1=A_2v_2. If A_2=A_1/2, then v_2=(A_1/A_2)v_1=2v_1: speed doubles.
Final answer: Doubles.

Question 11

Working: At equal height, p+(1/2)rho v^2=constant. If speed rises, the kinetic term rises, so static pressure must fall: p_2<p_1.
Final answer: Lower.

Question 12

Working: F=pA=5000(0.02)=100 N.
Final answer: 100 N.

Question 13

Working: For a liquid drop Delta p=2S/r=2(0.06)/0.003=0.12/0.003=40 Pa.
Final answer: 40 Pa.

Question 14

Working: A liquid drop has one liquid-air surface, so Delta p=2S/r. A soap bubble has inner and outer surfaces, so the two contributions give Delta p=4S/r, twice the drop's value at the same S and r.
Final answer: Bubble has two liquid-air surfaces; Deltap=4S/r.

Question 15

Working: A_1v_1=A_2v_2. Therefore v_2=(4 cm^2)(2 m/s)/(1 cm^2)=8 m/s.

Final answer: 8 m/s.

Question 16

Working: Bernoulli at equal height: p_1-p_2=(1/2)rho(v_2^2-v_1^2)=0.5(1000)(36-4)=500(32)=16,000 Pa.

Final answer: 16,000 Pa.

Question 17

Working: Stokes' drag F=6pi eta r v=6pi(1)(0.0005)(0.02)=1.884x10^-4 N, approximately 1.88x10^-4 N.

Final answer: approx 1.88 x 10^-4 N.

Question 18

Working: Consider a stationary column of liquid of height h. Its weight creates a pressure increment Delta p=weight/area=rho Ahg/A=rho gh. Therefore pressure increases with depth.

Final answer: Hydrostatic balance gives p=p_0+rhogh due to weight of fluid above.

Question 19

Working: Use continuity: v_2=A_1v_1/A_2=(6 cm^2)(1.5 m/s)/(2 cm^2)=4.5 m/s.

Final answer: 4.5 m/s.

Question 20

Working: For a soap bubble, Delta p=4S/r=4(0.03)/0.002=0.12/0.002=60 Pa.

Final answer: Deltap=4S/r=60 Pa.